Oscilloscope Manual & Data

Model: FNIRSI DS0-152
Compiled by David Malawey 2025

This document contains:

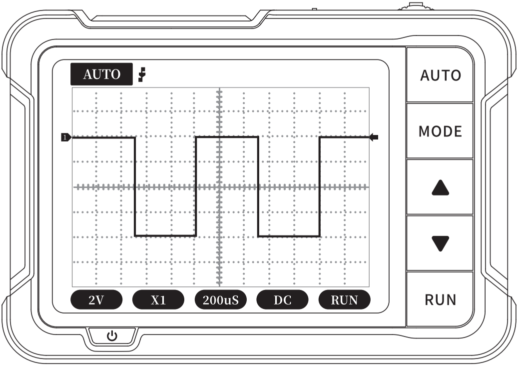
- Product package
- User Manual
- Probes Kit Manual

Product Package - oscilloscope from TikTok Shop, Sample sent for video promo 2025.04

FNIRSI 菲尼瑞斯

DSO152

DIGITAL OSCILLOSCOPE INSTRUCTION MANUAL



CATALOG

NOTICE TO USERS

●This manual introduces the usage method, precautions and related matters of the product in detail. Before using the product, please read the manual carefully in order to give full play to the best performance of the product.

●Do not use the device in a flammable or explosive environment.

●The used batteries replaced by the device and the discarded device cannot be disposed of together with domestic waste. Please dispose of them according to relevant national or local laws and regulations.

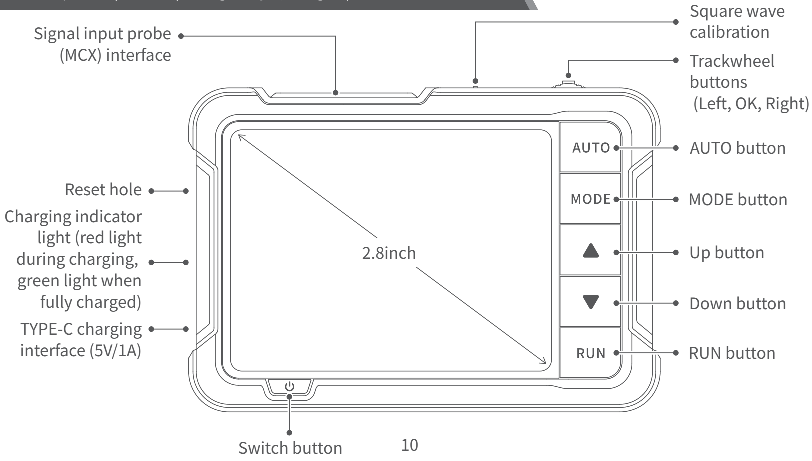
●When there is any quality problem with thedevice or you have any questions about the use of the it, please contact the online customer service or the manufacturer of "FNIRSI", and we will solve it for you in the first time.

1.PRODUCT INTRODUCTION

"DSO152" is a highly practical and cost-effective handheld oscilloscope launched by our company, which is aimed at the maintenance industry and the research education industry.The oscilloscope has a real-time sampling rate of 2.5MS/s, a bandwidth of 200KHz, and complete trigger functions (single, normal, and automatic). It can be used freely for both periodic analog signals and non-periodic digital signals, and can measure voltages up to ±400V. Equipped with an efficient one-key AUTO, it can display the measured waveform without cumbersome adjustments. Equipped with a

2.8-inch 320*240 resolution high-definition LCD screen.Built-in 1000mAh high-quality lithium battery, can be used continuously for about 4 hours after fully charged.

2.PANEL INTRODUCTION



3.KEY FUNCTION

BUTTON	OPERATION	FUNCTION
	Short press	Control parameters function selection
	Short press	Exit auto calibration (Auto calibration page)
	Long press	Enter the automatic calibration page
	Short press	Control parameters function selection
AUTO	Short press	Automatic adjustment (frequency below 45Hz cannot be calibrated correctly)
MODE	Short press	AUTO/Single/Normal switching
	Long press	Rising edge/falling edge switching

BUTTON	OPERATION	FUNCTION
▲	Short press	Parameter addition adjustment
▼	Short press	Parameter subtraction adjustment
RUN	Short press	Run/pause waveforms (other pages) Enter auto calibration (Auto calibration page)
	Long press	Show/close detailed parameters
⏻	Short press	OFF
	Long press	On

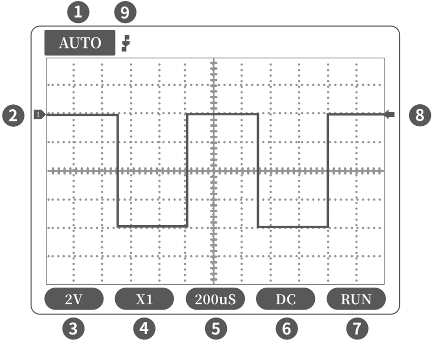
4.PARAMETER INDEX

Sampling rate	2.5MS/s
Bandwidth	200K
Vertical sensitivity	10mV/Div-10V/Div(Progress according to the 1-2-5 way)
Time base range	10μs/Div-50s/Div(Progress according to the 1-2-5 way)
Voltage range	X1:±40V (Vpp:80V)
	X10: ±400V (Vpp :800V)
Trigger method	Auto/Normal/Single
Coupling method	AC/DC
Display	2.8 inches/PPI:320*240
USB charging	5V/1A

Lithium battery capacity	1000mAh
Square wave calibration	Frequency: 1K Duty cycle: 50%
Size	99x68.3x19.5mm
Weight	100g

*Size and weight are measured manually, there may be some errors, please refer to the actual product.

5.SCREEN INSTRUCTIONS



①Trigger mode indicator icon, Auto means automatic trigger, Single means single trigger, Normal means normal trigger.

②Baseline indicator icon, this icon indicates the current position is OV voltage.

③Vertical sensitivity, which means the voltage represented by a large grid in the vertical direction.

④1X/10X mode indicator icon, this must be consistent with the 1X/10X switch setting on the probe handle, if the probe is in 1X mode, then the oscilloscope should also be set to 1X mode, 1X measures ±40V voltage, 10X measures ±400V voltage.

⑤Horizontal time base, indicating the length of time represented by a large grid in the horizontal direction.

⑥Input coupling indicator icon, AC means AC coupling, DC means DC coupling.

⑦Pause running indicator icon, RUN means running, STOP means pause.

⑧Trigger voltage indicator icon ⑨Trigger edge indicator icon

6.FIRMWARE UPGRADE

The device currently uses a USB analog U disk for firmware upgrade, and the upgrade steps are as follows:

①Press the "OK" button after pressing the power button so that to enter the U disk upgrade mode.

②Use the Type-C cable to connect the Type-C port on the board to the computer. At this time, the computer will pop up a U disk named "DSO BOOT".

③Pull the firmware into the U disk, and the firmware upgrade will complete.

NOTE

The firmware upgrade is only supported on the computer Windows 10 system.

7.PRECAUTIONS

●After receiving the device, please use it after fully charged.

●When using the oscilloscope, pay attention to the selection of the gear, the gear of the oscilloscope should be consistent with the gear of the probe.

●When measuring high voltage, it is forbidden to touch any metal part of the oscilloscope to avoid the risk of electric shock.

●Try not to perform a high voltage test while charging.

●When calibrating, need to unplug the BNC probe, or short the positive and negative poles of the probe.

●The USB firmware upgrade only supports WIN10 or above, and it is forbidden to drag in files other than the released firmware, otherwise it may cause unrecoverable consequences.

●Please use the voltage within the specification range of the manual for charging.

8.CONTACT US

Any FNIRSI'users with any questions who comes to contact us will have our promise to get a satisfactory solution + an Extra 6-Month Warranty to thanks for your support!

By the way, we have created an interesting community, welcome to contact FNIRSI staff to join our community.

Shenzhen FNIRSI Technology Co.,LTD.

Add.: West of Building C,Weida Industrial Park,Dalang Street,Longhua District,Shenzhen,Guangdong

E-mail: fnirsiofficial@gmail.com (Business)/ fnirsiofficialcs@gmail.com(Equipment service)

Tel: 0755-28020752 / +8613536884686

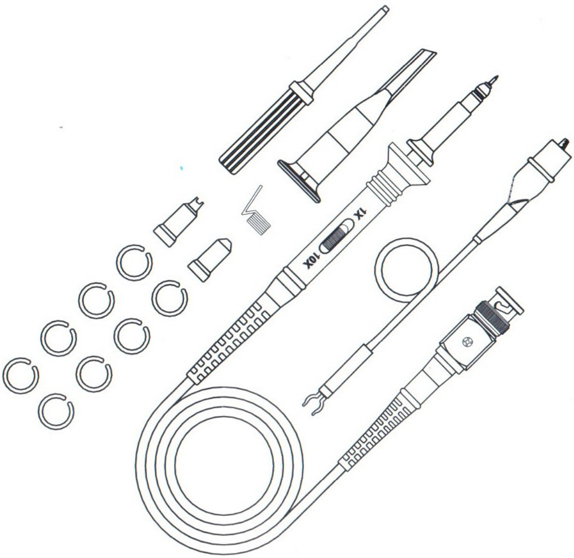
Web:www.fnirsi.cn

下载用户手册&应用软件
Download User manual&APP&Software

2025.06
Tiktok shop sample

User's Guide

- ☐ P6020 20MHz
- ☐ P6040 40MHz
- ☐ P6060 60MHz
- ☐ P6100 100MHz
- ☐ P6150 150MHz
- ☐ P6200 200MHz



P6000 1X&10X Oscilloscope Probe

Probe Characteristics

Model		P6020	P6040	P6060	P6100	P6150	P6200
Bandwidth	1X	35MHz			10MHz		
	10X	20MHz	40MHz	60MHz	100MHz	150MHz	200MHz
Rise time	1X	10ns			35ns		
	10X	17.5ns	8.75ns	5.8ns	3.5ns	2.3ns	1.75ns
Attenuation Ratio		1X&10X					
Input Resistance		1MΩ/10MΩ±2%					
Input Capacitance		1X:100-160pF			1X:60-120pF		
		10X:18-22pF			10X:13.5-16.5pF		
Max. Input Voltage		1X:150V RMS CAT II&10X:300V RMS CAT II					
Compensation Range		15~45pF					
Operation Temperature		-10℃~+55℃					
Humidity		-40℃ or below ≤90% +41℃ to +50℃ ≤60%					
Altitude		Operating 3000m Nonoperating 15000m					
Cable Length		110±2cm					
Weight		About 55g					
Safety		IEC61010-031					

Accessory Kit

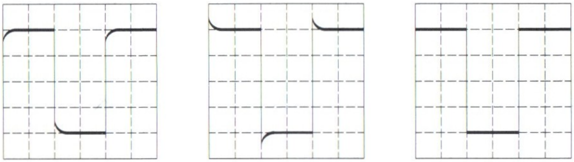
Item	Description	1PCS/	2PCS/
1	Retractable Hook Tip	1	2
2	Adjustment Tool	1	
3	Locating Sleeve	2	
4	Marker Rings	8	
5	Ground Lead	1	2
6	Ground Spring(Above 100MHz)	1	

Note:
content of this document are subject to change without notice.

Version 2024 4.0

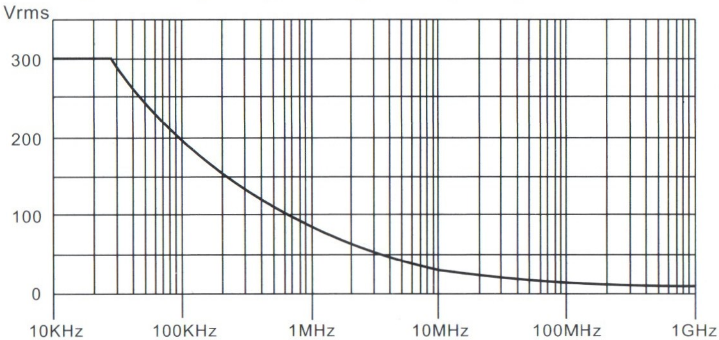
Frequency Compensation

Before taking any measurements using a probe, first check the compensation of the probe and adjust it to match the channel inputs. Most oscilloscopes have a square wave reference signal available at a terminal on the front panel used to compensate the probe. Connect the probe to the signal source on your oscilloscope. Set the probe to 10X position. Adjust trimmer until seeing flat-top square wave on the display.



Incorrect Incorrect Correct

Voltage vs Frequency Rating Curve(RMS)



CAT II IEC measurement category II is for measurements performed on circuits directly connected to low voltage installations. Examples are measurements on household appliances, portable tools and similar equipment.

⧈ Equipment fully protected by DOUBLE INSULATTION or REINFORCED INSULATION.

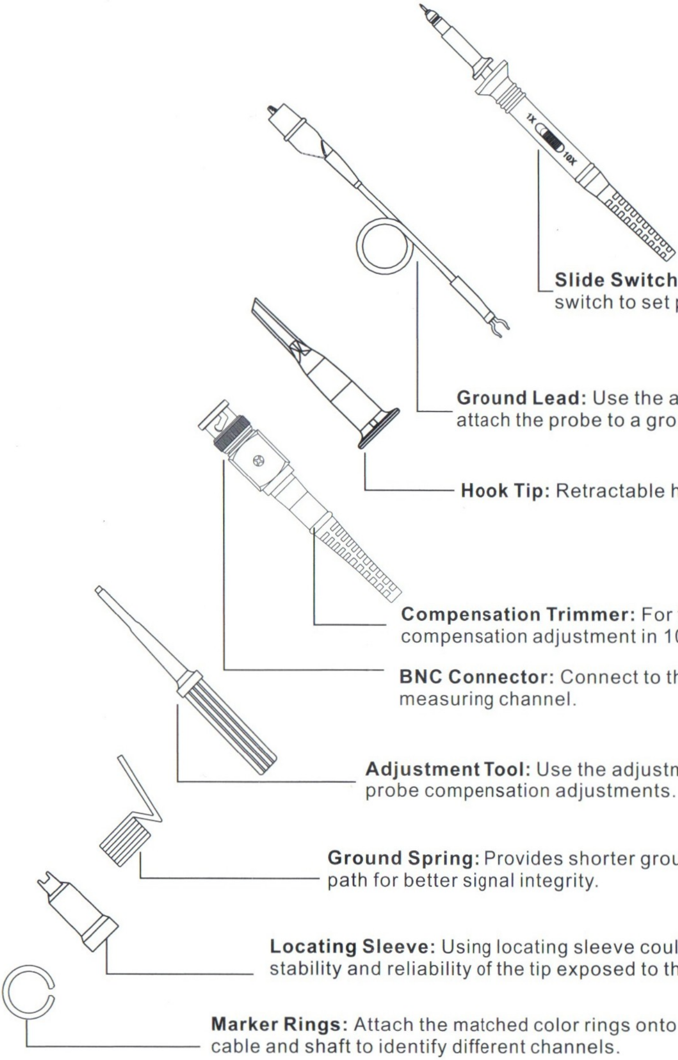
⚠ Review this user manual carefully to avoid any personal injury or damage to this product and any product connected to it. To avoid potential hazards, use this product only as specified.

⚠ The measurement category of a combination of a PROBE ASSEMBLY and an accessory (an auxiliary to the measurement) is the lower of the measurement categories of the PROBE ASSEMBLY and of the accessory.

⚠ Follow the instructions to use the PROBE ASSEMBLY will impair or ruin its self-owned protection function.

Accessories and Features

Probe is provided with several accessories designed to make probing and measurement simpler. Please take a moment to familiarize yourself with these accessories and their uses.



Slide Switch: Use the slide switch to set probe attenuation.

Ground Lead: Use the alligator clip to attach the probe to a ground reference.

Hook Tip: Retractable hook tip.

Compensation Trimmer: For frequency compensation adjustment in 10 X mode.

BNC Connector: Connect to the oscilloscope measuring channel.

Adjustment Tool: Use the adjustment tool for probe compensation adjustments.

Ground Spring: Provides shorter grounding path for better signal integrity.

Locating Sleeve: Using locating sleeve could ensure the stability and reliability of the tip exposed to the test point.

Marker Rings: Attach the matched color rings onto the probe cable and shaft to identify different channels.